Remarks on the continuity of optimal transport maps

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Abstract

We provide a proof of the continuity of an optimal transport map from a measure in $L^\infty(\mathbb{R}^2)$ to a measure $\nu \in \mathscr{P}_2(\mathbb{R}^2)$ with compact convex support dominating an absolutely continuous measure with density bounded away from 0. The support of the first measure is not constrained and the second measure might be singular. The argument is quite elementary and does not rely on PDEs.

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AI statement. No AI was used to prepare this note. But don't worry, my human slop is as unreadable.

The optimal transport problem between measures μ, ν consists in finding the assignment of the mass of μ to the mass of ν that minimises a travel cost. Monge started to investigate the situation in which this assignment does not split mass, in the sense that each particle x seen by μ is sent to a unique particle $f(x)$ seen by ν. This problems has by now a quite satisfactory partial answer [Bre91; McC01; Gig11] based on the measure that μ gives to certain small sets. Different points of view conclude that this situation is the most common one: [BPS25, Theorem 1.6] shows that the set of (μ, ν) for which one optimal plan splits mass is of first category in the sense of Baire (small), [VRS09] constructs a "Gaussian measure on measures" for which the set of problematic measures μ is negligible, and [Tak11] showed that when restricting to Gaussian measures, the problematic μ's are isometric to a set of codimension one.

In comparison, the cases where the optimal map is regular seem quite few. In this note, regular means continuous. As for existence, it is easier to shift the question: instead of wondering which pairs (μ, ν) admit a continuous optimal map, one rather looks for conditions on one marginal (here ν) under which continuity holds for *any* other marginal. Caffarelli [Caf92] showed that the relevant assumption is the convexity of the support of ν. Precisely, if μ, ν are absolutely continuous measures with compact convex support, and densities bounded away from 0 and ∞, then the unique optimal transport map f from μ to ν is continuous. The convexity enters when showing that f is the gradient of an Alexandrov solution of the Monge-Ampère equation $\det(\nabla^2 u) = \mu/\nu(\nabla u)$. One knows by [Bre91] that f is the gradient of a convex function u, and the Alexandrov measure $\det(\nabla^2 u)$ can always be computed; however, in presence of discontinuities of f, it might have singular parts that the measure $\mu/\nu(\nabla u)$ does not record. Once the equivalence is established, the regularity properties follow from that of the Monge-Ampère equation, as detailed in [DP13].

A new area of research was opened by Ma, Trudinger and Wang [MTW05] in introducing a condition on the cost under which regularity results hold. For such costs, smooth measures with smooth suitably convex supports admits smooth optimal transport maps. Loeper [Loe09] showed that the MTW condition is is fact necessary, and studied a "strengthened" version under which continuity of maps can be obtained even for measures that would not admit a density. Precisely, one considers measures μ, ν such that $\mu(\mathscr{B}(x,r))$ decays faster than r^{d-1} for any x, and $\nu(\mathscr{B}(x,r)) \geqslant \lambda \mathcal{L}(\mathscr{B}(x,r))$ for some positive $\lambda > 0$. Then, under suitable strict convexity assumptions on the supports, [Loe09, Theorem 3.4] yields continuity of the optimal transport map with a quantitative control on the modulus of continuity. Manifolds in which a similar result holds are said to have the *transport continuity property* (TCP) in the terminology of [FR09]. TCP is known to hold for spheres and the cost $(x, y) \mapsto -\log(|x - y|)$ [Loe11], as well as for some small C^4 deformations of such spaces [FR09; LV10; DG10]. It seems to be still open in $\mathbb{R}^2$. Lastly me mention that [FGP05] could obtain regularity results for the cost $c(x, y) = |x - y|$ in dimension 2 (even thought the latter does not satisfy the MTW condition) by decomposing the transport plan on transport rays.

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Statements. This note is motivated by an example of Mérigot suggesting that the optimal transport map between a Hausdorff measure on the Von Koch curve and the Lebesgue measure on $[0,1]^2$ is continuous. This precise example is still out of reach, but we could obtain slight improvements on the known results in $\mathbb{R}^2$, fitting between the regularity given by Caffarelli and the transport continuity property. We start by the situation for $d = 1$, which, although being very simple, is hardly found in the literature. The rest of the paper settles in dimension $d = 2$. We do not see strong obstructions to treating higher dimension similarly.

Theorem 1. *Let $\mu, \nu \in \mathscr{P}_2(\mathbb{R}^2)$ be probability measures such that*

(h_μ). $\mu \leqslant \Lambda \mathcal{L}$ for some $\Lambda \in (0,\infty)$, where $\mathcal{L}$ is the Lebesgue measure on $\mathbb{R}^2$,

(h_ν). ν has convex compact support, and $\nu \geqslant \lambda \mathcal{L} \llcorner \operatorname{supp} \nu$ for some $\lambda > 0$.

Then the unique optimal transport plan between μ and ν is induced by a continuous map $f \in C(\operatorname{supp} \mu; \mathbb{R}^2)$.

Some comments on (h_μ)-(h_ν). The assumption that the support of ν is compact can be relaxed, but one should then allow f to take infinite values, or define it only on a subset of supp μ. We discuss this in dimension one in Corollary 2.2 and assume supp ν to be compact in dimension 2. In general, the assumption that $\nu \geqslant \lambda \mathcal{L} \llcorner \operatorname{supp} \nu$ cannot be removed: a counterexample is given in Section 4 based on the Monge-Ampère construction of [Wan95]. The sharp assumption here stays to be found. On the other hand, the assumption of convexity of supp ν is the most relevant one, and cannot be removed (see Proposition 2.1 for $d = 1$). The assumption of boundedness of the density of μ is more annoying, since it does not intervene in the statements in curved manifolds. We do not know how to relax it so far.

Strategy. The proof is neither particularly beautiful nor very informative, but it relies on completely elementary arguments. We do not exploit the relation with the Monge-Ampère equation. Instead, we assume by contradiction that the optimal map is discontinuous at some point x, i.e. there are two different points y_0, y_1 such that (x, y_0) and (x, y_1) belong to the support of the optimal plan. Then we construct a triangle $T_n \subset \operatorname{conv}\{y_0, y_1, y_n\} \subset \operatorname{supp} \nu$ for y_n converging to a point in $[y_0, y_1]$. The assumption (h_ν) gives a lower bound on the mass that this triangle receives. On the other hand, using monotonicity, we can bound the region of points $x' \in \operatorname{supp} \mu$ that are sent to T_n by the optimal transport. The assumption (h_μ) gives an upper bound on the mass that μ puts in this region. Comparing the two in (3), we get a contradiction.

To apply the argument, we need a *nondegenerate* triangle T_n, i.e. with positive Lebesgue measure – otherwise $\nu(T_n)$ could be 0. In addition, we need to relate the point $y_n \in \operatorname{supp} \nu$ to a point $x_n \in \operatorname{supp} \mu$ converging to x in a suitable way, i.e. with $x_n - x$ becoming increasingly orthogonal to $y_1 - y_0$ (assumption (h3) in Lemma 3.3). This control on the convergence allows to close the above estimate on the side of μ. Getting it is easy when y_n converges to a point lying strictly between y_0 and y_1 (Lemma 3.4), but more intricate otherwise. The difficulty there is linked to the fact that (a) there might not exist an optimal map between ν and μ, and (b) even if it exists, it might not be continuous. However these cases correspond to the situation in which f, going from μ to ν, is more and more constant. This plays in the direction of continuity, and assuming to be in this situation, one gets additional properties that allow to conclude.

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1 Preliminaries

Notations. The open ball with center x and radius r is denoted $\mathscr{B}(x,r)$. The closure of a set A is denoted $\overline{A}$, its complement A^c, and its convex hull conv A. We often use the short notation $x+\mathbb{R}\cdot v$ to denote the set of points that can be written as $x+\alpha v$ for some $\alpha\in\mathbb{R}$. The function $\mathbb{1}_A$ is the indicator of A, i.e. $\mathbb{1}_A(x)=1$ if $x\in A$ and 0 otherwise, while $\mathbb{1}_A^\infty$ is the convex indicator of A, i.e. $\mathbb{1}_A^\infty(x)=0$ if $x\in A$, and $+\infty$ otherwise.

The set of continuous functions from X to Y is denoted $C(X,Y)$, and L^p spaces associated to a measure μ by $L^p_\mu(X:Y)$ with $Y=\mathbb{R}^d$. By a *set-valued function* or a *multifunction*, we mean an application $F:X\rightrightarrows Y$ such that $F(x)$ is a subset of Y. F is upper semi-continuous (in the set-valued sense) if for any open set $\mathcal{O}\supset F(x)$, there exists $r>0$ such that $y\in\mathscr{B}(x,r)$ implies $F(y)\subset\mathcal{O}$ [AF90]. If φ is convex, $\partial_x\varphi$ denotes its subdifferential at x, i.e. the set of vectors w such that $\varphi(y)\geqslant\varphi(x)+\langle y-x,w\rangle$ for any point y. The Fenchel transform of φ is defined as $\varphi^*(y):=\sup_x\langle x,y\rangle-\varphi(x)$.

Wasserstein space. Let μ be a probability measure on $\mathbb{R}^d$. We say that $\mu\in\mathscr{P}_2(\mathbb{R}^d)$ if $\int_{x\in\mathbb{R}^d}|x|^2d\mu<\infty$. The support of μ is the (closed) set of points $x\in\mathbb{R}^d$ such that $\mu(\mathscr{B}(x,r))>0$ for any $r>0$. If $A\subset\mathbb{R}^d$ is measurable, the restricted measure $\mu\llcorner A$ is defined by $(\mu\llcorner A)(B)=\mu(A\cap B)$. If $f:\mathbb{R}^d\to X$ is a measurable map with values in a topological space X, the pushforward measure $f_\#\mu$ is defined by $(f_\#\mu)(A)=\mu(f^{-1}(A))$ for any measurable $A\subset X$. In particular, if $\eta=\eta(dx,dy)$ is a measure on $X\times Y$, we denote $\pi_x:X\times Y\to X$ the projection on the first component, and refer to the pushforward $\pi_{1\#}\eta$ as the first marginal of η.

If $\mu,\nu\in\mathscr{P}_2(\mathbb{R}^d)$, we define the set of *transport plans* from μ to ν as

$$\Gamma(\mu,\nu):=\left\{\eta=\eta(dx,dy)\in\mathscr{P}_2(\mathbb{R}^d\times\mathbb{R}^d)\;\middle|\;\pi_{x\#}\eta=\mu\text{ and }\pi_{y\#}\eta=\nu\right\}.$$

The Wasserstein distance $W(\cdot,\cdot)$ between μ and ν is then given by

$$W^2(\mu,\nu):=\inf_{\eta\in\Gamma(\mu,\nu)}\int_{(x,y)\in(\mathbb{R}^d)^2}|x-y|^2d\eta(x,y).$$

In our setting, the infimum is attained on a set that we denote $\Gamma_o(\mu,\nu)$. The reader is referred to [San15] for an introduction to the theory of optimal transport. We use in particular the specificities of dimension one, in which the set $\Gamma_o(\mu,\nu)$ always reduces to a singleton that is explicitly given in function of the quantile maps (Theorem 2.9 and Definition 2.3 of [San15]). In addition, we extensively use the cyclical monotonicity of the support of optimal plans: if $\eta\in\Gamma_o(\mu,\nu)$, then for any sequence $(x_n,y_n)_{n=1}^N\subset\operatorname{supp}\eta$, there holds

$$\sum_{n=1}^N\left|x_n-y_n\right|^2\leqslant\sum_{n=1}^N\left|x_n-y_{n+1}\right|^2,\qquad\text{or equivalently}\qquad\sum_{n=1}^N\langle x_n,y_n\rangle\geqslant\sum_{n=1}^N\langle x_n,y_{n+1}\rangle.$$

By convention $y_{N+1}=y_1$. By a theorem of Rockafellar, cyclical monotonicity is equivalent to inclusion in the subdifferential of a proper, lower semi-continuous convex function [Roc70, Theorem 24.8]. The identification of these functions for optimal transport can be found in [Vil09, Theorem 5.10].

Transport-regularity. The terminology of transport-regularity was coined by [Jui11] to refer to measures $\mu\in\mathscr{P}_2(\mathbb{R}^d)$ such that for any other measure $\nu\in\mathscr{P}_2(\mathbb{R}^d)$, the set $\Gamma_o(\mu,\nu)$ reduces to a singleton η, and $\eta=(id,f)_\#\mu$ for some $f\in L^2_\mu(\mathbb{R}^d;\mathbb{R}^d)$. In this case we say that η is induced by the map f. The Brenier-McCann-Gigli theorem characterizes transport-regular measures as the ones giving 0 mass to sets of non-differentiability of convex functions, and by [Zaj79], this is equivalent to the fact that $\mu(A)=0$ whenever A can be covered by graph of a difference of convex functions from $\mathbb{R}^{d-1}\to\mathbb{R}$, up to a permutation of variables. We refer to [Gig11] for the precise statement. In this note, we only use that transport-regular measures in dimension 1 are exactly non-atomic measures, and that in general, they give 0 mass to lines.

Lastly we record the following elementary lemmas on supports.

Lemma 1.1. *Let $\eta\in\mathscr{P}_2((\mathbb{R}^d)^2)$ have first marginal μ. If μ is concentrated on a set $A\subset\mathbb{R}^d$, then $G_A:=\left\{(x,y)\in\operatorname{supp}\eta\;\middle|\;x\in A\right\}$ is dense in* $\operatorname{supp}\eta$.

Proof. Let $(x,y)\in\operatorname{supp}\eta$ be in the complement of $\overline{G_A}$. Then $\mathscr{B}((x,y),\varepsilon)\subset G_A^c$ for some $\varepsilon>0$. By definition of the support, $0<\eta(\mathscr{B}((x,y),\varepsilon))\leqslant\eta(G_A^c)=\eta(A^c\times\mathbb{R}^d)=\mu(A^c)=0$, a contradiction. □

Lemma 1.2 (Projection of the support)**.** *Let $\mu, \nu \in \mathscr{P}_2(\mathbb{R}^d)$. For any $\eta \in \Gamma(\mu,\nu)$,*

$$\pi_1(\operatorname{supp}\eta) := \left\{ x \;\middle|\; \left(\{x\}\times\mathbb{R}^d\right) \cap \operatorname{supp}\eta \neq \emptyset \right\} \subset \operatorname{supp}\mu.$$

If moreover ν is compactly supported, then equality holds.

Proof. Let first $x \notin \operatorname{supp}\mu$. Then there exists $\varepsilon > 0$ such that $0 = \mu(\mathscr{B}(x,\varepsilon)) = \eta\left(\mathscr{B}(x,\varepsilon)\times\mathbb{R}^d\right)$. Therefore there cannot be any point of the support of η in the open set $\mathscr{B}(x,\varepsilon)\times\mathbb{R}^d$, and $x \notin \pi_1 \operatorname{supp}\eta$. This shows that $\pi_1 \operatorname{supp}\eta \subset \operatorname{supp}\mu$, and symmetrically, that $\pi_2 \operatorname{supp}\eta \subset \operatorname{supp}\nu$.

If ν has compact support, let $x \in \operatorname{supp}\mu$ and $(\varepsilon_n)_n$ be a decreasing sequence converging to 0. Then

$$\eta\left(\overline{\mathscr{B}}(x,\varepsilon_n)\times\operatorname{supp}\nu\right) = \eta\left(\overline{\mathscr{B}}(x,\varepsilon_n)\times\mathbb{R}^d\right) = \mu(\overline{\mathscr{B}}(x,\varepsilon_n)) > 0 \qquad \text{for each } n\in\mathbb{N}.$$

Consequently $\left(\operatorname{supp}\eta \cap (\overline{\mathscr{B}}(x,\varepsilon_n)\times\operatorname{supp}\nu)\right)_{n\in\mathbb{N}}$ is a nested sequence of nonempty compact sets, which must have nonempty intersection. Any point in the intersection belongs to $\operatorname{supp}\eta \cap (\{x\}\times\mathbb{R}^d)$, so that $x \in \pi_1\operatorname{supp}\eta$. This shows the reverse inclusion. □

2 The case of dimension one

In dimension one, optimal plans are always unique [San15, Theorem 2.9]. We prove the following result.

Proposition 2.1. *Assume $\nu \in \mathscr{P}_2(\mathbb{R})$ has compact convex support. If there exists an optimal map between μ and ν, then there exists a continuous optimal map $f \in C(\operatorname{supp}\mu;\mathbb{R})$ between μ and ν.*

Proof. Denote η the optimal transport plan from μ to ν. By Lemma 1.2, the set-valued function

$$G(x) := \left\{\, y\in\mathbb{R} \;\middle|\; (x,y)\in\operatorname{supp}\eta \,\right\}$$

has nonempty values if and only if $x \in \operatorname{supp}\mu$. Moreover $G(x)$ is closed and contained in the compact $\operatorname{supp}\nu$. We prove that for each $x \in \operatorname{supp}\mu$, the set $G(x)$ contains exactly one point.

Assume by contradiction that $y_0 := \min G(x)$ is strictly inferior to $y_1 := \max G(x)$. Both points y_0 and y_1 belong to the support of ν, which is convex by assumption, so that $[y_0,y_1] \subset \operatorname{supp}\nu$. Let $y \in (y_0,y_1)$. By Lemma 1.2, there exists $z \in \operatorname{supp}\mu$ such that $(z,y) \in \operatorname{supp}\eta$. As the support of η is monotone, there holds

$$(y-y_0)(z-x) \geqslant 0 \qquad \text{and} \qquad (y_1-y)(x-z) \geqslant 0,$$

from which we deduce that $z = x$. Hence

$$0 < \nu((y_0,y_1)) = \eta\left(\mathbb{R}\times(y_0,y_1)\right) = \eta(\{x\}\times(y_0,y_1)) \leqslant \mu(\{x\}),$$

and x is an atom of μ. By assumption, there exists $g \in L^2_\mu(\mathbb{R};\mathbb{R})$ inducing the plan η. The value of g is unambiguously defined at the atom x, and $(x,g(x))$ belongs to $\operatorname{supp}\eta$, so that $g(x) \in G(x) \subset [y_0,y_1]$. Let I be an open interval contained in $(y_0,y_1)\setminus\{g(x)\}$. Since $I \subset \operatorname{supp}\nu$, there holds $\nu(I) > 0$. On the other hand,

$$0 < \nu(I) = \eta(I\times\mathbb{R}) = \eta\left([I\times(-\infty,x)]\cap\operatorname{supp}\eta\right) + \eta\left([I\cap(x,+\infty)]\cap\operatorname{supp}\eta\right). \tag{1}$$

If (x',y') is a point of the support of η such that $x' < x$, then $(y_0-y')(x-x') \geqslant 0$ by monotonicity. Consequently $y' \leqslant y_0$ does not belong to I. Similarly, if $x' > x$, then $y' \geqslant y_1$ does not belong to I. Therefore the rightmost term of (1) vanishes, a contradiction.

Define now $f : \operatorname{supp}\mu \to \mathbb{R}$ such that $G(x) = \{f(x)\}$. Then $\operatorname{supp}\eta = \operatorname{graph} f$ is closed. As f is also valued in the compact $\operatorname{supp}\eta$, it must be continuous, providing the desired map. □

This statement is quite sharp: if supp ν is not convex, we can find a transport-regular μ such that any optimal map is discontinuous. Indeed, there must be a nontrivial interval $[a,b] \subset (\operatorname{supp}\nu)^c$ such that both sets $\operatorname{supp}\nu \cap (-\infty, a)$ and $\operatorname{supp}\nu \cap (b,\infty)$ are nonempty. Let $\mu := (\mathcal{L} \llcorner [a,b])/|b-a|$. By the Brenier theorem, the optimal plan between μ and ν is induced by a nondecreasing map, say $f : \mathbb{R} \to \mathbb{R}$. Let $x \in [a,b]$ be such that $F_\mu(x) = \mu([a,x]) = \nu((-\infty,a]) = F_\nu(a)$. Then

$$y < x \quad \Longrightarrow \quad F_\mu(y) < F_\mu(x) = F_\nu(a) \quad \Longrightarrow \quad f(y) = F_\nu^{[-1]}(F_\mu(y)) \leqslant a,$$

and similarly, $y > x$ implies $f(y) \geqslant b$. Hence f is discontinuous at x.

We state the corresponding result when the support of ν is not compact. In this case, the map f might be unbounded on the extremities of the support of μ, as for $\mu = \mathcal{L} \llcorner [0,1]$ and ν a Gaussian measure.

Corollary 2.2. *Let $\nu \in \mathscr{P}_2(\mathbb{R})$ have convex support. If there exists an optimal map between $\mu \in \mathscr{P}_2(\mathbb{R})$ and ν, then there exists $f \in C(\operatorname{supp}\mu \setminus \{\inf\operatorname{supp}\mu, \sup\operatorname{supp}\mu\}; \mathbb{R})$ optimal between μ to ν.*

Proof. Let $g : \mathbb{R} \to (-1,1)$ be a continuous increasing bijection, for instance $g(x) = \tan^{-1}(x)$. Then $g_\# \nu$ has compact convex support. By Proposition 2.1, there exists $f \in C(\operatorname{supp}\mu; \mathbb{R})$ inducing the optimal transport plan between μ and $g_\#\nu$. If $f(x) = 0$ for some $x \in \mathbb{R}$, then $x = \inf\operatorname{supp}\mu$ and x cannot be an atom of μ. Similarly, $f(x) = 1$ only if $x = \sup\operatorname{supp}\mu$ is not an atom. Therefore $g^{-1} \circ f$ is defined and continuous on $\operatorname{supp}\mu \setminus \{\inf\operatorname{supp}\mu, \sup\operatorname{supp}\mu\}$, and satisfies by definition $(g^{-1}\circ f)_\#\mu = g_\#^{-1} f_\# \mu = g_\#^{-1} g_\# \nu = \nu$. As $g^{-1}\circ f$ is monotone by composition, it must induce the optimal plan between μ and ν [San15, Lemma 2.8]. □

3 Results in dimension two

We start by the following proposition.

Proposition 3.1. *Let $\mu, \nu \in \mathscr{P}_2(\mathbb{R}^2)$ be such that*

- $\operatorname{supp}\mu = \mathbb{R}^2$ *and* $\mu \leqslant \Lambda \mathcal{L}$ *for some finite* $\Lambda > 0$,
- $\operatorname{supp}\nu$ *is convex, compact, has positive Lebesgue measure and* $\nu \geqslant \lambda \mathcal{L} \llcorner \operatorname{supp}\nu$ *for some* $\lambda > 0$.

Then the unique optimal transport plan $\eta \in \Gamma_o(\mu,\nu)$ is induced by a map $f \in C\left(\mathbb{R}^2; \mathbb{R}^2\right)$.

The assumptions that μ has full support and $\mathcal{L}(\operatorname{supp}\nu) > 0$ will be removed in Section 3.3. As in the case of dimension 1, we introduce the multifunction $G : \mathbb{R}^2 \rightrightarrows \mathbb{R}^2$ as

$$G(x) := \left\{ y \in \mathbb{R}^2 \;\middle|\; (x,y) \in \operatorname{supp}\eta \right\}.$$

The assumptions of Proposition 3.1 imply that G has nonempty compact images. It is also upper semi-continuous in the set-valued sense. Hence Proposition 3.1 will follow if we show that $G(x)$ contains exactly one point for each $x \in \mathbb{R}^2$.

3.1 From anything to at most two points

Lemma 3.2. *Assume that μ is non-atomic and that ν has convex support. The set* $\operatorname{conv} G(x)$ *has empty interior for any $x \in \operatorname{supp}\mu$.*

Proof. Assume by contradiction that for some $x \in \operatorname{supp}\mu$, the set $\operatorname{conv} G(x)$ contains an open ball $\mathscr{B}(y_0, \varepsilon)$. Then $\nu(\operatorname{int}\operatorname{conv} G(x)) > 0$ by our assumptions. Consider now $(x', y') \in \operatorname{supp}\eta$ with $x' \neq x$. By monotonicity, for any $y \in G(x)$, there holds $\langle y' - y, x' - x\rangle \geqslant 0$. Introducing $\pm\langle y_0, x' - x\rangle$ and taking a sup over $y \in G(x)$,

$$\langle y' - y_0, x' - x\rangle \geqslant \sup_{y \in G(x)} \langle y - y_0, x' - x\rangle = \sup_{y \in \operatorname{conv} G(x)} \langle y - y_0, x' - x\rangle \geqslant \varepsilon |x' - x| > 0. \tag{2}$$

If y' were in the interior of $\operatorname{conv} G(x)$, there would exist $r > 1$ such that $y_0 + r(y' - y_0) \in \operatorname{conv} G(x)$, and then

$$\sup_{y \in \operatorname{conv} G(x)} \langle y - y_0, x' - x\rangle \geqslant \langle y_0 + r(y' - y_0) - y_0, x' - x\rangle = r\,\langle y' - y_0, x' - x\rangle > \langle y' - y_0, x' - x\rangle,$$

against (2). We just proved that whenever $(x', y') \in \operatorname{supp} \eta$ with $x' \notin x$, there holds $y' \notin \operatorname{int}\operatorname{conv} G(x)$. Equivalently, $\operatorname{supp} \eta \cap (\mathbb{R}^d \times \operatorname{int}\operatorname{conv} G(x)) \subset \{x\} \times \operatorname{int}\operatorname{conv} G(x)$. which implies that

$$0 < \nu(\operatorname{int}\operatorname{conv} G(x)) = \eta\Big(\mathbb{R}^d \times \operatorname{int}\operatorname{conv} G(x)\Big) \leqslant \eta(\{x\} \times \operatorname{int}\operatorname{conv} G(x)) \leqslant \mu(\{x\}),$$

and x would be an atom of μ, against the assumption. □

The following lemma is the main argument in the sequel.

Lemma 3.3. *Consider the assumptions of Proposition 3.1. Let $y_0, y_1 \in \mathbb{R}^2$ and $(x_n, y_n)_{n\in\mathbb{N}}$ be such that*

(h1). $\lim_n x_n = x$ *for some* $x \in \mathbb{R}^2$ *with* $\{y_0, y_1\} \subset G(x) \subset [y_0, y_1]$,

(h2). $\operatorname{dist}(y_n, y_0 + \mathbb{R}\cdot(y_1 - y_0)) > 0$ *for any* n,

(h3). there exists $c \in (0,\infty)$ *and* $n_0 \in \mathbb{N}$ *such that*

$$|\langle x_n - x, y_1 - y_0\rangle| \leqslant c|x - x_n| \operatorname{dist}(y_n, y_0 + \mathbb{R}\cdot(y_1 - y_0)) \qquad \forall\, n \geqslant n_0.$$

Then $y_0 = y_1$.

Proof. Assume by contradiction that $y_0 \neq y_1$. The strategy is to build a sequence of triangles T_n in the support of ν, and to prove that the total mass that μ can send to T_n decreases as $o(\nu(T_n))$. For convenience, denote $L_{01} := y_0 + \mathbb{R}\cdot(y_1 - y_0)$ the line passing through y_0 and y_1, and $y_t := (1-t)y_0 + ty_1$ for each $t \in [0,1]$. Up to extraction, we can fix a unit vector v orthogonal to $y_1 - y_0$ such that $\langle y_n - y_0, v\rangle > 0$ for any n.

Since $(y_n)_n \subset \operatorname{supp} \nu$, which is compact, we can extract a further subsequence (still denoted $(y_n)_n$) converging to some $y \in \operatorname{supp} \nu$. The limit y belongs to $G(x) \subset [y_0, y_1]$ by upper semi-continuity of G. We assume that $y \in [y_0, (y_0 + y_1)/2]$, the other case being symmetric. For n large enough, the segment $[y_n, y_{3/4}]$ intersects the half-line $y_{2/3} + \mathbb{R}^+ \cdot v$ at a unique point $\overline{y}_n$ (see Figure 1). Denote

$$T_n := \operatorname{int}\operatorname{conv}\{y_{2/3}, y_{3/4}, \overline{y}_n\}$$

the interior of the triangle formed by $y_{2/3}, y_{3/4}$ and $\overline{y}_n$. Then $T_n \subset \operatorname{conv}\{y_0, y_1, y_n\} \subset \operatorname{supp} \nu$ by convexity of the support of ν. The rest of the argument will focus on the region

$$S_n := \{\, x' \in \mathbb{R}^2 \;|\; \exists y' \in T_n \text{ such that } (x', y') \in \operatorname{supp} \eta \,\}.$$

S_n is measurable by standard arguments. We claim that $\mathcal{L}(S_n) \leqslant \overline{c}|x - x_n|^2 \mathcal{L}(T_n)$ for some constant $\overline{c}$ independent of n. Let us finish the proof assuming this bound to hold. Using that $\nu \geqslant \lambda\mathcal{L}\llcorner \operatorname{supp} \nu$ and $\mu \leqslant \Lambda\mathcal{L}$,

$$\lambda\mathcal{L}(T_n) \leqslant \nu(T_n) = \eta(\mathbb{R}^2 \times T_n) = \eta(S_n \times T_n) \leqslant \eta(S_n \times \mathbb{R}^2) = \mu(S_n) \leqslant \Lambda\mathcal{L}(S_n) \leqslant \Lambda\overline{c}|x - x_n|^2 \mathcal{L}(T_n). \tag{3}$$

As $x_n \to x$, this is absurd for large n. Hence $y_0 = y_1$.

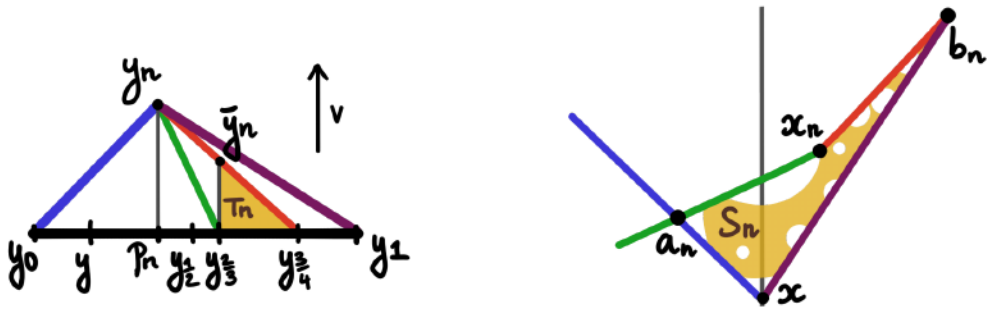


Figure 1: Left: definition of T_n. Right: construction of a bound on S_n.

Covering of S_n by two triangles. Let $(x', y') \in \operatorname{supp} \eta$ such that $y' \in \operatorname{conv}\{y_0, y_1, y_n\} \setminus [y_0, y_1]$. By the monotonicity of the support of η applied with the points $(x, y_0), (x, y_1) \in \operatorname{supp} \eta$, there holds

$$\langle y' - y_0, x' - x\rangle \geqslant 0, \qquad \text{and} \qquad \langle y' - y_1, x' - x\rangle \geqslant 0.$$

The vector $y' - y_0$ belongs to the open cone enclosed by the half-lines $\mathbb{R}^+ \cdot (y_n - y_0)$ and $\mathbb{R}^+ \cdot (y_1 - y_0)$. Therefore the first inequality constrains x' to belong to the obtuse open cone enclosed by the semi-lines $x + \mathbb{R}^+ \cdot R(y_n - y_0)$ and $x + \mathbb{R}^+ \cdot R^{-1}(y_1 - y_0)$, for R a rotation by an angle $\pm\pi/2$. Similarly, the second inequality forces x' to belong to the obtuse open cone enclosed by the half-lines $x + \mathbb{R}^+ \cdot R(y_0 - y_1)$ and $x + \mathbb{R}^+ \cdot R^{-1}(y_n - y_1)$. The intersection of these cones is the acute cone $\mathcal{C}$ enclosed by the two half-lines $x + \mathbb{R}^+ \cdot R(y_n - y_0)$ and $x + \mathbb{R}^+ \cdot R^{-1}(y_n - y_1)$. In particular, x_n and S_n are contained in $\mathcal{C}$.

On the other hand, let $x' \in S_n$ and $y' \in T_n \cap G(x')$. Monotonicity with the point (x_n, y_n) yields

$$0 \leqslant \langle x' - x_n, y' - y_n \rangle . \tag{4}$$

By the same computations as before, x' is constrained to belong to the obtuse cone enclosed by the half-lines $x_n + \mathbb{R}^+ \cdot R^{-1}(y_{3/4} - y_n)$ and $x_n + \mathbb{R}^+ \cdot R(y_{2/3} - y_n)$. The vectors $y_n - y_0$ and $y_{2/3} - y_0$ are not collinear by assumption, hence the two lines $x + \mathbb{R} \cdot R(y_n - y_0)$ and $x_n + \mathbb{R} \cdot R(y_{2/3} - y_n)$ intersect at a unique point a_n. For the same reason, $x + \mathbb{R} \cdot R^{-1}(y_n - y_1)$ and $x_n + \mathbb{R} \cdot R^{-1}(y_{3/4} - y_n)$ intersect at a unique point b_n. Taking into account that x_n belongs to $\mathcal{C}$, we obtain a cover of S_n by the two closed triangles $\mathrm{conv}\{x, x_n, a_n\}$ and $\mathrm{conv}\{x, x_n, b_n\}$. The area of the region S_n can therefore be bounded as

$$\mathcal{L}(S_n) \leqslant \mathcal{L}\left(\mathrm{conv}\{x_n, x, a_n\}\right) + \mathcal{L}(\mathrm{conv}\{x_n, x, b_n\}) = \frac{1}{2}\left[|\det(x_n - x, a_n - x)| + |\det(x_n - x, b_n - x)|\right]. \tag{5}$$

Bound on the size of each triangle. We now estimate the right hand-side of (5). By definition of the point a_n, there exist $\alpha, \beta \in \mathbb{R}$ such that

$$a_n = x + \alpha R(y_n - y_0) = x_n + \beta R(y_{2/3} - y_n).$$

Subtracting x and taking a scalar product with $y_{2/3} - y_n$, we get $\alpha \langle R(y_n - y_0), y_{2/3} - y_n \rangle = \langle x_n - x, y_{2/3} - y_n \rangle$. On the one hand, recognizing the area of a triangle and using the base×height/2 formula,

$$|\langle R(y_n - y_0), y_{2/3} - y_n \rangle| = |\det(y_0 - y_n, y_{2/3} - y_n)| = 2\mathcal{L}(\mathrm{conv}\{y_0, y_{2/3}, y_n\}) = |y_{2/3} - y_0| \,\mathrm{dist}(y_n, L_{01}).$$

On the other hand, let p_n be the projection of y_n on $L_{01} = y_0 + \mathbb{R} \cdot (y_1 - y_0)$, so that $|y_n - p_n| = \mathrm{dist}(y_n, L_{01})$. The vector $y_{2/3} - p_n$ is equal to $r_n(y_1 - y_0)$ for some $r_n \in [1/6, 1]$, so that using the assumption (h3),

$$\left|\langle x_n - x, y_{2/3} - y_n \rangle\right| \leqslant \left|\langle x_n - x, y_{2/3} - p_n \rangle\right| + \left|\langle x_n - x, p_n - y_n \rangle\right| \leqslant (c+1)|x_n - x| \,\mathrm{dist}(y_n, L_{01}). \tag{6}$$

Therefore, recalling that we defined $a_n = x + \alpha R(y_n - y_0)$,

$$|\det(x_n - x, a_n - x)| = |\alpha| |\det(x_n - x, R(y_n - y_0))| \leqslant \frac{(c+1)|x_n - x| \,\mathrm{dist}(y_n, L_{01})}{|y_{2/3} - y_0| \,\mathrm{dist}(y_n, L_{01})} |\langle x_n - x, y_0 - y_n \rangle|.$$

The term $\mathrm{dist}(y_n, L_{01})$ cancels. The rightmost factor can be estimated exactly as in (6) with y_0 in place of $y_{2/3}$, and we obtain the bound $|\det(x_n - x, a_n - x)| \leqslant (c+1)^2 |x_n - x|^2 \,\mathrm{dist}(y_n, L_{01}) / |y_{2/3} - y_0|$. Now, the triangles $\mathrm{conv}\{p_n, y_{3/4}, y_n\}$ and $T_n = \mathrm{conv}\{y_{2/3}, y_{3/4}, \overline{y_n}\}$ are rescalings of each other, so that

$$\frac{\mathrm{dist}(y_n, L_{01}) |p_n - y_{3/4}|}{2} = \mathcal{L}(\mathrm{conv}\{p_n, y_{3/4}, y_n\}) = \mathcal{L}\left(\frac{|p_n - y_{3/4}|}{|y_{2/3} - y_{3/4}|} T_n\right) = \left(\frac{|p_n - y_{3/4}|}{|y_{2/3} - y_{3/4}|}\right)^2 \mathcal{L}(T_n).$$

As p_n converges towards $y \in [y_0, y_{1/2}]$, there holds $0 < |p_n - y_{3/4}| \leqslant |y_1 - y_0|$ for n large enough. In conclusion

$$|\det(x_n - x, a_n - x)| \leqslant \frac{(c+1)^2 |x_n - x|^2 \,\mathrm{dist}(y_n, L_{01})}{|y_{2/3} - y_0|} \leqslant \frac{2(c+1)^2 |y_1 - y_0|}{|y_{2/3} - y_0| |y_{2/3} - y_{3/4}|^2} \times |x_n - x|^2 \times \mathcal{L}(T_n).$$

The same computation yields the same bound for $|\det(x_n - x, b_n - x)|$, with $|y_{2/3} - y_0|$ replaced by $|y_{3/4} - y_1|$. Hence there exists $\overline{c}$ such that for sufficiently large n, $\mathcal{L}(S_n) \leqslant \overline{c} |x_n - x|^2 \mathcal{L}(T_n)$, as promised. □

We can now show that the set-valued application G has images containing at most two points.

Lemma 3.4. *Under the assumptions of Proposition 3.1, $G(x)$ contains at most two elements for each $x \in \mathbb{R}^2$.*

Proof. Assume that there exists $x \in \mathbb{R}^2$ with three distinct elements. By Lemma 3.2, $G(x)$ is contained in a subspace of dimension one, and it is compact; hence we can find $y_0 \neq y_1$ and $y \in (y_0, y_1)$ such that $\{y_0, y, y_1\} \subset G(x) \subset [y_0, y_1]$. We construct a sequence satisfying the assumptions of Lemma 3.3.

Let $B := x + \mathbb{R} \cdot (y_1 - y_0)^\perp$. Since $\mu(B) = 0$, we can apply Lemma 1.1 to find a sequence $(x_n, y_n)_n \to (x, y)$ with $x_n \notin B$ and $y_n \in G(x_n)$. In particular, the projection p_n of y_n on $y_0 + \mathbb{R} \cdot (y_1 - y_0)$ converges towards y. Let n_0 be large enough so that $p_n \in (y_0, y_1)$. For such n, monotonicity yields

$$0 \leqslant \langle x_n - x, y_n - y_0 \rangle = \langle x_n - x, y_n - p_n \rangle + \langle x_n - x, p_n - y_0 \rangle, \qquad \text{and} \tag{7}$$

$$0 \leqslant \langle x_n - x, y_n - y_1 \rangle = \langle x_n - x, y_n - p_n \rangle + \langle x_n - x, p_n - y_1 \rangle. \tag{8}$$

As we chose $x_n \notin B$, the terms $\langle x_n - x, p_n - y_0 \rangle$ and $\langle x_n - x, p_n - y_1 \rangle$ are nonzero with opposite signs. Hence $\langle x_n - x, y_n - p_n \rangle$ cannot be zero, which shows that $|y_n - p_n| = \operatorname{dist}(y_n, y_0 + \mathbb{R} \cdot (y_1 - y_0)) > 0$. Thus $(x_n, y_n)_n$ satisfies the assumptions (h1)-(h2) of Lemma 3.3.

To prove (h3), observe that (7)-(8) yield $\langle x_n - x, p_n - y_i \rangle \geqslant -|x_n - x||y_n - p_n|$ for $i \in \{0, 1\}$. Let $r_n > 0$ be such that $p_n - y_1 = -r_n(p_n - y_0)$, with r_n converging to some $r > 0$. We deduce that

$$-|x_n - x||p_n - y_n| \leqslant \langle x_n - x, p_n - y_1 \rangle = -r_n \langle x_n - x, p_n - y_0 \rangle \leqslant r_n |x_n - x||p_n - y_n|,$$

hence that $\big|\langle x_n - x, p_n - y_1 \rangle\big| \leqslant \max(1, r_n)|p_n - y_n||x_n - x|$ for each n. On the other hand $y_0 - y_1 = s_n(p_n - y_1)$ for some $s_n \to s \in (0, \infty)$. Therefore for sufficiently large n,

$$\big|\langle x_n - x, y_0 - y_1 \rangle\big| \leqslant s_n \max(1, r_n)|x_n - x||p_n - y_n| \leqslant 2s \max(1, r)|x_n - x| \operatorname{dist}\big(y_n, y_0 + \mathbb{R} \cdot (y_1 - y_0)\big).$$

This shows the third assumption of Lemma 3.3, and we deduce that $y_0 = y_1$, a contradiction. □

3.2 From two points to one point

The reduction from two to one point turn out to be more involved. As in Lemma 3.4, the strategy is to construct a sequence on which to apply Lemma 3.3. However, we cannot anymore assume that the sequence converges to a point of the *interior* of the segment containing $G(x)$. This was the argument allowing us to obtain (h2) and (h3) through monotonicity. Hence we go through several modifications of an initial sequence to recover these assumptions.

Underlyingly, we are treating the cases in which ν would put mass on points or segments. In these cases, the optimal plan η might concentrate mass on the singletons $\{y_0\}$, $\{y_1\}$ or the segment $[y_0, y_1]$, and the optimal map inducing it might not be injective. Intuitively, the optimal map becomes "more constant", hence goes in the favourable direction for continuity. The successive modifications that we perform might be interpreted as eliminating these easier cases until coming back to the situation in which the map is injective, and then apply the argument of Lemma 3.3.

3.2.1 Preliminary results

We first show a form of rigidity of the multi-function G. The set G is contained in the sub-differential of a convex function by cyclical monotonicity [Vil09, Theorem 5.10], but if μ does not see singularities of convex functions, then it has the additional property of lying on the boundary of the subdifferential.

Lemma 3.5. *Assume that $\mu \in \mathscr{P}_2(\mathbb{R}^2)$ is transport-regular, and let $\eta \in \Gamma_o(\mu, \nu)$ for some $\nu \in \mathscr{P}_2(\mathbb{R}^2)$. Let $\varphi : \mathbb{R}^2 \to \mathbb{R}$ be a Lipschitz convex function such that $\eta(\partial\varphi) = 1$, where $\partial\varphi = \big\{(x, y) \;\big|\; y \in \partial_x\varphi\big\}$. Then*

$$D_x\varphi(w) := \lim_{h \searrow 0} \frac{\varphi(x + hw) - \varphi(x)}{h} = \sup_{y \in G(x)} \langle y, w \rangle \tag{9}$$

for any $x \in \mathbb{R}^2$ and $w \in \mathbb{R}^2$.

Note that (9) is not implied by the cyclical monotonicity of G, which only gives an inclusion into $\partial\varphi$. For instance, in $\mathbb{R}$, the set $G = \mathbb{R}_*^- \times \{-1\} \cup \{(0, 0)\} \cup \mathbb{R}_*^+ \times \{1\}$ is cyclically monotone, but does not satisfy (9) for any convex function φ.

Proof. As ∂S is closed, one has $\operatorname{supp}\eta \subset \partial S$. Hence the inequality $D_x\varphi(w) = \sup_{y\in\partial_x\varphi}\langle y, w\rangle \geqslant \sup_{y\in G(x)}\langle y, w\rangle$ holds everywhere. Conversely, the assumption of transport-regularity implies that $\partial_x\varphi$ reduces to a singleton μ-almost everywhere. Equivalently, μ is concentrated on a set $A \subset \mathbb{R}^2$ such that φ is differentiable at any $x \in A$. For such x, the set $G(x)$ is equal to $\{\nabla_x\varphi\}$ (since it is nonempty), and the equality in (9) follows.

We show that equality holds for $x \notin A$. Let then $w \in \mathbb{R}^2$. By compactness of $G(x)$, there exists $y^* \in G(x)$ realizing $\sup_{y\in G(x)}\langle y, w\rangle$. Lemma 1.1 furnishes a sequence $(x_n, y_n)_n \subset \operatorname{supp}\eta$ converging to (x, y^*) and such that $x_n \in A$ for all n. Using that $x \mapsto D_x\varphi(w)$ is upper semi-continuous, since it is the nonincreasing limit of continuous functions, we get

$$\sup_{y\in G(x)}\langle y, w\rangle = \langle y^*, w\rangle = \lim_{n\to\infty}\langle y_n, w\rangle = \lim_{n\to\infty}\langle\nabla_{x_n}\varphi, w\rangle = \limsup_{n\to\infty} D_{x_n}\varphi(w) \geqslant D_x\varphi(w) = \sup_{y\in\partial_x\varphi}\langle y, w\rangle\,.$$

Therefore equality holds everywhere. □

Lemma 3.6 (Rigidity)**.** *Consider the assumptions of Proposition 3.1. Assume that $x, x' \in \mathbb{R}^2$ are such that $G(x) = \{y_0, y_1\}$, $G(x') = \{y_0', y_1'\}$ with $y_1 - y_0$, $y_1' - y_0'$ nonzero vectors both orthogonal to $x' - x$. Then $G(x) = G(x') = G((1-t)x + tx')$ for any $t \in (0,1)$.*

Proof. We can rename the points so that $\langle y_1' - y_0', y_1 - y_0\rangle > 0$. Denote $x_t := (1-t)x + tx'$ for $t \in (0,1)$. By Lemma 3.4, the set $G(x_t)$ is given by $\{y_0^t, y_1^t\}$ for (possibly equal) points y_0^t, y_1^t such that $\langle y_1^t - y_0^t, y_1 - y_0\rangle \geqslant 0$. By monotonicity,

$$0 \leqslant \langle y_0^t - y_0, x_t - x\rangle\,, \qquad \text{and} \qquad 0 \leqslant \langle y_0^t - y_0', x_t - x'\rangle = -\frac{1-t}{t}\langle y_0^t - y_0, x_t - x\rangle\,.$$

Here we used that $\langle y_0' - y_0, x_t - x\rangle = 0$. Therefore $\langle y_0^t - y_0, x_t - x\rangle = 0$ for any $t \in [0,1]$.

Let now φ be a Lipschitz convex function as in Lemma 3.5. For any $t \in (0,1)$, there holds

$$\lim_{h\searrow 0}\frac{\varphi(x_{t+h}) - \varphi(x_t)}{h} = D_{x_t}\varphi(x' - x) = \sup_{y\in\partial_{x_t}\varphi}\langle y, x' - x\rangle = \sup_{y\in G(x_t)}\langle y, x' - x\rangle = \langle y_0, x' - x\rangle\,.$$

The rightmost term does not depend on t. Integrating, we get that $\varphi(x_t) = \varphi(x) + \langle y_0, x_t - x\rangle$ for any $t \in [0,1]$. We show that y_1^t is constant with respect to $t \in (0,1)$, or equivalently, that $\langle y_1^t - y_1^s, y_1 - y_0\rangle = 0$ for any $s, t \in (0,1)$. Let then $s, t \in [0,1]$ and $w := y_1 - y_0$. Our choice of names yields that $D_{x_t}\varphi(w) = \langle w, y_1^t\rangle$ for any $t \in [0,1]$. By convexity of φ, and the fact that $y_1^s \in \partial_{x_s}\varphi$,

$$\begin{aligned}\varphi(x_t + hw) &\geqslant \varphi(x_s) + \langle y_1^s, x_t + hw - x_s\rangle = \varphi(x_t) + \langle y_0, x_s - x_t\rangle + \langle y_1^s, x_t + hw - x_s\rangle\\ &= \varphi(x_t) + \underbrace{\langle y_0 - y_1^s, x_s - x_t\rangle}_{=0} + h\langle y_1^s, w\rangle\,.\end{aligned}$$

Passing to the limit as $h \searrow 0$, we get $D_{x_t}\varphi(w) = \langle w, y_1^t\rangle \geqslant \langle y_1^s, w\rangle$. As s, t are arbitrary, we conclude that $y_1^t \equiv y_1$. The symmetric reasoning with $w = y_0 - y_1$ yields that $y_0^t \equiv y_0$, and we conclude. □

3.2.2 Proof of Proposition 3.1

The statement that we want to prove is the following. If μ has full support and satisfies $\mu \leqslant \Lambda\mathcal{L}$ for some finite Λ, and if ν has compact convex support with the lower bound $\nu \geqslant \lambda\mathcal{L}\llcorner\operatorname{supp}\nu$ (and $\mathcal{L}(\operatorname{supp}\nu) > 0$), then each $G(x)$ contains exactly one point.

Proof of Proposition 3.1. Assume that there exists $\overline{x} \in \mathbb{R}^2$ such that $G(\overline{x})$ contains more than one element. By Lemma 3.4, $G(\overline{x}) = \{y_0, y_1\}$ for $y_0 \neq y_1$. As in there, we construct a sequence on which Lemma 3.3 applies.

Modification of the base point. We first modify the point $\overline{x}$. Let R be the matrix of the rotation by $\pi/2$. Since the segment $[y_0, y_1]$ is contained in $\operatorname{supp}\nu$, which is a convex set of positive Lebesgue measure, one of the two open half-planes $H_\pm := \{y \in \mathbb{R}^2 \mid \langle y - y_0, \pm R(y_1 - y_0)\rangle > 0\}$ must intersect $\operatorname{supp}\nu$. Up to renaming the points, let us assume that it is the case for H^+, and denote $v := R(y_1 - y_0)$. Let then

$$K := \{\, x' \in \overline{x} + \mathbb{R}^+ \cdot v \;\big|\; G(x') \subset y_0 + \mathbb{R}\cdot(y_1 - y_0)\,\}\,. \tag{10}$$

Then K is nonempty (since $\overline{x} \in K$), closed (since G is upper semi-continuous), and by Lemma 3.6, the set K is convex with $G(x') = G(x) = \{y_0, y_1\}$ for any $x' \in K$.

Moreover, as H^+ intersects the support of ν, which is convex, there must be $\varepsilon > 0$ small enough so that $y^\varepsilon := y_{1/2} + \varepsilon\nu \in \operatorname{supp}\nu$. Let $x^\varepsilon \in \mathbb{R}^2$ be such that $y^\varepsilon \in G(x^\varepsilon)$. For any $x' \in K$, one has

$$0 \leqslant \langle x^\varepsilon - x', y^\varepsilon - y_0\rangle = \langle x^\varepsilon - x', \varepsilon\nu\rangle + \langle x^\varepsilon - x', y_{1/2} - y_0\rangle, \qquad \text{and}$$
$$0 \leqslant \langle x^\varepsilon - x', y^\varepsilon - y_1\rangle = \langle x^\varepsilon - x', \varepsilon\nu\rangle + \langle x^\varepsilon - x', y_{1/2} - y_1\rangle = \langle x^\varepsilon - x', \varepsilon\nu\rangle - \langle x^\varepsilon - x', y_{1/2} - y_0\rangle.$$

Hence $\langle x^\varepsilon - x', \varepsilon\nu\rangle \geqslant |\langle x^\varepsilon - x', y_{1/2} - y_0\rangle| \geqslant 0$ for each $x' \in K$. Recalling that x' is of the form $\overline{x} + t\nu$ for $t \in \mathbb{R}^+$, we deduce that K is compact. For the rest of the proof, we fix $x \in K$ as a point of maximum of the function $\langle x', \nu\rangle$ over $x' \in K$. In particular, $G(x) = G(\overline{x}) = \{y_0, y_1\}$.

Suitable approximations of y_0, y_1. By upper semi-continuity of G, there exists $\varepsilon > 0$ such that $G(x') \subset \mathscr{B}(y_0, |y_0 - y_1|/3) \cup \mathscr{B}(y_1, |y_0 - y_1|/3)$ for any $x' \in \mathscr{B}(x, \varepsilon)$. Let $s_n \to 0$ be decreasing. For each n, denote

$$x_n^0 := x + s_n((y_0 - y_1) + \nu), \qquad \text{and} \qquad x_n^1 := x + s_n((y_1 - y_0) + \nu).$$

We drop the first terms of the sequence to ensure that $x_n^0, x_n^1 \in \mathscr{B}(x, \varepsilon)$. Then, for any $y \in G(x_n^0)$, there holds

$$0 \leqslant \langle x_n^0 - x, y - y_0\rangle = \langle x_n^0 - x, y - y_1\rangle + \langle x_n^0 - x, y_1 - y_0\rangle \leqslant s_n\sqrt{2}|y_1 - y_0||y - y_1| - s_n|y_0 - y_1|^2,$$

so that $|y - y_1| \geqslant \frac{|y_0 - y_1|}{\sqrt{2}} > \frac{|y_0 - y_1|}{3}$. Therefore $y \in \mathscr{B}(y_0, |y_0 - y_1|/3)$. Similarly, $G(x_n^1) \subset \mathscr{B}(y_1, |y_1 - y_0|/3)$.

Construction of the sequence. We now show that some point x' of the segment $[x_n^0, x_n^1]$ must have a set $G(x')$ intersecting both balls centered at y_0, y_1. Indeed, for a fixed n, consider the function

$$\tau \in [0,1] \mapsto G(x_n^\tau), \qquad \text{where } x_n^\tau := (1 - \tau)x_n^0 + \tau x_n^1.$$

This is a multifunction with nonempty compact values, and it is upper semi-continuous by restriction. Let

$$A_0 := \left\{\tau \in [0,1] \;\middle|\; G(x_n^\tau) \cap \overline{\mathscr{B}}(y_0, |y_1 - y_0|/3) \neq \emptyset\right\}, \qquad A_1 := \left\{\tau \in [0,1] \;\middle|\; G(x_n^\tau) \cap \overline{\mathscr{B}}(y_1, |y_1 - y_0|/3) \neq \emptyset\right\}.$$

Both sets A_0, A_1 are closed, nonempty (since $0 \in A_0$ and $1 \in A_1$), and convex. Indeed, if $0 \leqslant \tau < \tau'$ and $\tau' \in A_0$, then for any $y \in G(x_n^\tau)$ and $y' \in G(x_n^{\tau'})$, there holds

$$\langle y - y', x_n^\tau - x_n^{\tau'}\rangle = (\tau' - \tau)\langle y - y', x_n^0 - x_n^1\rangle = 2s_n(\tau' - \tau)\langle y - y', y_0 - y_1\rangle \geqslant 0.$$

This is only possible if $y \in \mathscr{B}(y_0, |y_1 - y_0|/3)$. The same reasoning shows that A_1 is also an interval. Now, $A_0 \cup A_1$ covers $[0,1]$, hence they must intersect at a point τ_n. Denote $x_n := x_n^{\tau_n}$, which is such that $G(x_n) = \{y_n^0, y_n^1\}$. We now check the assumptions of Lemma 3.3.

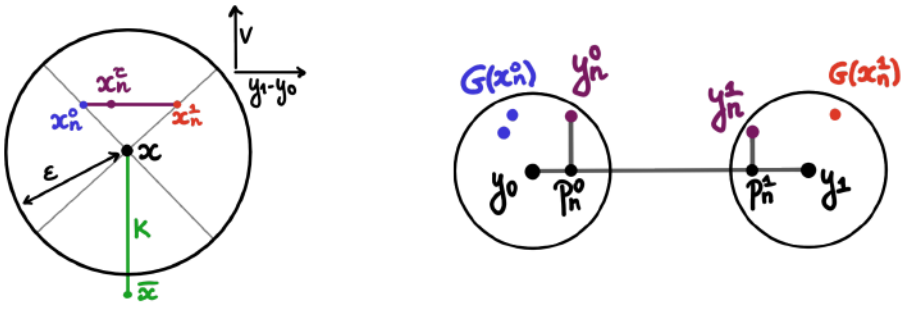


Figure 2: Notations.

Satisfaction of (h1)-(h2). First, the sequence x_n converges to x, since it is a convex combination of x_n^0 and x_n^1 which both converge to x. Secondly, we claim that for each n, at least one point of $G(x_n)$ does not belong to $y_0 + \mathbb{R} \cdot (y_1 - y_0)$. Otherwise, it would hold that

$$\langle y_n^0 - y_1, x_n - x\rangle \geqslant 0 \qquad \text{and} \qquad \langle y_n^1 - y_0, x_n - x\rangle \geqslant 0,$$

with $y_n^0 - y_1$ and $y_n^1 - y_0$ nonzero vectors collinear to $y_1 - y_0$ with opposite signs, hence $\langle x_n - x, y_1 - y_0\rangle = 0$. However in this case x_n would belong to the set K defined in (10). Since there holds

$$\langle x_n - x, v\rangle = (1-\tau_n)\langle x_n^0 - x, v\rangle + \tau_n \langle x_n^1 - x, v\rangle = s_n |v|^2 > 0,$$

this would contradict our choice of x as the point of K maximizing the scalar product with v. Hence for each n, $G(x_n) \not\subset y_0 + \mathbb{R}\cdot(y_1 - y_0)$, which is equivalent to

$$\max_{i\in\{0,1\}} |\langle y_n^i - y_0, v\rangle| = \max_{i\in\{0,1\}} \operatorname{dist}(y_n^i, y_0 + \mathbb{R}\cdot(y_1 - y_0)) > 0. \tag{11}$$

For each n, let $i_n \in \{0,1\}$ be an index realizing the maximum. There must be a subsequence $(i_{n_k})_{k\in\mathbb{N}}$ for which $i_{n_k} \equiv i^*$ is constant. We choose this subsequence and drop the notation $(n_k)_k$. For the rest of the proof we assume that $i^* = 0$, the case $i^* = 1$ being symmetric. Hence, by now, the maximum in (11) is reached for $i = 0$ for any n.

Satisfaction of (h3). Since y_n^0 converges to $y_0 \notin (y_0, y_1)$, we cannot use the same argument as in Lemma 3.4 to show the third assumption. However we can use the point y_n^1. Denote p_n^0, p_n^1 the projections of y_n^0, y_n^1 on the line $y_0 + \mathbb{R}\cdot(y_1 - y_0)$. There holds

$$\begin{aligned} 0 \leqslant \langle x_n - x, y_0^n - y_1\rangle &= \langle x_n - x, y_0^n - p_0^n\rangle + \langle x_n - x, p_0^n - y_1\rangle, \qquad \text{and} \\ 0 \leqslant \langle x_n - x, y_1^n - y_0\rangle &= \langle x_n - x, y_1^n - p_1^n\rangle + \langle x_n - x, p_1^n - y_0\rangle. \end{aligned}$$

Since y_n^0 converges to y_0, there holds $y_0 - y_1 = s_n(p_0^n - y_1)$ for some $s_n \to 1$. This yields

$$\langle x_n - x, y_0 - y_1\rangle = s_n\langle x_n - x, p_0^n - y_1\rangle \geqslant -s_n\langle x_n - x, y_0^n - p_0^n\rangle \geqslant -s_n|x_n - x|\operatorname{dist}(y_0^n, y_0 + \mathbb{R}\cdot(y_1 - y_0)). \tag{12}$$

Similarly, for $r_n \to 1$ such that $y_1 - y_0 = r_n(p_1^n - y_0)$,

$$\langle x_n - x, y_1 - y_0\rangle \geqslant -r_n|x_n - x|\operatorname{dist}(y_1^n, y_0 + \mathbb{R}\cdot(y_1 - y_0)) \geqslant -r_n|x_n - x|\operatorname{dist}(y_0^n, y_0 + \mathbb{R}\cdot(y_1 - y_0)). \tag{13}$$

The last inequality comes from the fact that (11) is maximized by $i = 0$. Consequently (12) and (13) yield

$$\left|\langle x_n - x, y_0 - y_1\rangle\right| \leqslant \max(s_n, r_n)|x_n - x|\operatorname{dist}\left(y_0^n, y_0 + \mathbb{R}\cdot(y_1 - y_0)\right),$$

with $\max(s_n, r_n) \leqslant 2$ for n large enough. Therefore the three assumptions of Lemma 3.3 are satisfied by the sequence $(x_n, y_0^n)_{n\in\mathbb{N}}$. As a result, $y_0 = y_1$, contradicting the assumption. □

3.3 Proof of Theorem 1

We finish the proof of Theorem 1 by removing the assumption of full support of μ and nondegenerate support of ν in Proposition 3.1. The case where ν has a support of dimension 1 can actually be treated separately with a stronger conclusion.

Lemma 3.7. *Assume that* $\operatorname{supp}\nu \subset \mathbb{R}^2$ *is compact, convex and has 0 Lebesgue measure. If* μ *is transport-regular, then the optimal transport map* $f \in L^2_\mu$ *between* μ *and* ν *extends continuously on* $\operatorname{supp}\mu$.

Proof. Let $L \subset \mathbb{R}^2$ be a line containing $\operatorname{supp}\nu$, and $p : \mathbb{R}^2 \to L$ the orthogonal projection on L. By a gluing argument, there holds $W^2(\mu,\nu) = W^2(\mu, p_\#\mu) + W^2(p_\#\mu, \nu)$. Now, $p_\#\mu$ is non-atomic. Indeed, $p_\#\mu(\{x\}) = \mu(p^{-1}(\{x\}))$. The set $p^{-1}(\{x\})$ is a line, and in particular the set of non-differentiability of the convex function $y \mapsto d(y, p^{-1}(\{x\}))$, so that μ gives 0 mass to it. Therefore, by Brenier-McCann-Gigli [Gig11], there exists an optimal map $\widetilde{f} \in L^2_{p_\#\mu}(L; L)$ between $p_\#\mu$ and ν. By Proposition 2.1, $\widetilde{f}$ is continuous on $\operatorname{supp} p_\#\mu = p(\operatorname{supp}\mu)$. The composition $\widetilde{f}\circ p$ is a continuous map pushing μ on ν, and there holds

$$\int |\widetilde{f}(p(x)) - x|^2 d\mu = \int |\widetilde{f}(p(x)) - p(x)|^2 d\mu + \int |p(x) - x|^2 d\mu = W^2(\nu, p_\#\mu) + W^2(p_\#\mu, \mu) = W^2(\nu,\mu).$$

Hence $f \in C(\operatorname{supp}\mu; \mathbb{R}^2)$ induces the (unique) optimal plan from μ to ν. □

Lastly we remove the assumption that μ has full support.

Proof of Theorem 1. Consider $\mu \in \mathscr{P}_2(\mathbb{R}^2)$ such that $\mu \leqslant \Lambda\mathcal{L}$ for some $\Lambda \in (0,\infty)$, and $\nu \in \mathscr{P}_2(\mathbb{R}^2)$ with compact convex support such that $\nu \geqslant \lambda\mathcal{L}\llcorner \operatorname{supp}\nu$ for some $\lambda > 0$. If $\mathcal{L}(\operatorname{supp}\nu) = 0$, Lemma 3.7 provides a continuous optimal map between μ and ν. Otherwise, let $\eta \in \Gamma_o(\mu,\nu)$ be the unique optimal transport plan.

By [Vil09, Theorem 5.10], there exists a convex function $\varphi : \mathbb{R}^2 \to \mathbb{R}$ such that $\operatorname{supp}\eta \subset \partial\varphi$. We modify it so that the image of the subdifferential is equal to $\operatorname{supp}\nu$. Consider the convex Lipschitz function

$$\widetilde{\varphi}(x) := \sup_{y\in\operatorname{supp}\nu} \langle x,y\rangle - \varphi^*(y) = \sup_y \langle x,y\rangle - (\varphi^* + \mathbb{I}^\infty_{\operatorname{supp}\nu})(y).$$

Since $\operatorname{supp}\nu$ is convex, so is the proper, lower semi-continuous function $\varphi^* + \mathbb{I}^\infty_{\operatorname{supp}\nu}$, and therefore $(\widetilde{\varphi})^* = \varphi^* + \mathbb{I}^\infty_{\operatorname{supp}\nu}$. There holds $\widetilde{\varphi}(x) \leqslant \varphi(x)$ for all points. Moreover, if $y \in \partial_x\varphi \cap \operatorname{supp}\nu$ for some $x \in \operatorname{supp}\mu$, then

$$\varphi(x) + \varphi^*(y) = \langle x,y\rangle \qquad \Longrightarrow \qquad \widetilde{\varphi}(x) + (\varphi^* + \mathbb{I}^\infty_{\operatorname{supp}\nu})(y) \leqslant \langle x,y\rangle \qquad \Longrightarrow \qquad y \in \partial_x\widetilde{\varphi}.$$

Hence $\operatorname{supp}\eta \subset \partial\widetilde{\varphi}$. Note that $y \in \partial_x\widetilde{\varphi}$ implies that $y \in \operatorname{supp}\nu$; indeed, $y \in \partial_x\widetilde{\varphi}$ if and only if $x \in \partial_y(\widetilde{\varphi})^*$, and the subdifferential of $\widetilde{\varphi}$ is empty at any point at which $\widetilde{\varphi}$ is infinite.

Let $\rho : \mathbb{R}^2 \to \mathbb{R}^+$ be a smooth strictly positive function with integral 1. The measure

$$\widetilde{\mu} := \frac{\mu + \rho\,\mathbb{1}_{\operatorname{supp}\mu^c}\mathcal{L}}{2}$$

has full support. Take a measurable selection $f : \mathbb{R}^2 \to \mathbb{R}^2$ of the subdifferential of $\widetilde{\varphi}$, for instance given by the measurable selection theorem [AF90, Theorem 8.3.1]. Consider

$$\widetilde{\eta} := \frac{\eta + (id,f)_\#\rho\,\mathbb{1}_{\operatorname{supp}\mu^c}\mathcal{L}}{2}.$$

Then $\widetilde{\eta}$ is still supported on $\partial\widetilde{\varphi}$, hence an optimal transport plan between its first marginal $\widetilde{\mu}$, and its second marginal $\widetilde{\nu}$. The latter measure is equal to $\frac{1}{2}\nu + \frac{1}{2}\omega$ for $\omega = f_\#\rho\,\mathbb{1}_{\operatorname{supp}\mu^c}\mathcal{L}$. Since f is valued in $\operatorname{supp}\nu$ there holds $\operatorname{supp}\widetilde{\nu} = \operatorname{supp}\nu$. Moreover $\widetilde{\nu} \geqslant \frac{\lambda}{2}\mathcal{L}\llcorner\operatorname{supp}\widetilde{\nu}$. Applying Proposition 3.1, there exists a continuous map $f \in C(\mathbb{R}^2;\mathbb{R}^2)$ such that $\widetilde{\eta} = (id,f)_\#\widetilde{\mu}$. By restriction on $\operatorname{supp}\mu \times \mathbb{R}^2$, this implies that $\eta = (id,f)_\#\mu$ for the same continuous map f, and we conclude. □

4 Counterexample of Wang when $\lambda = 0$

Here we take inspiration from the Example 3 in [Wan95] to provide an explicit counterexample to regularity when $\lambda = 0$. Consider the function

$$\psi(x,y) = y^2\varphi(x/y^2), \qquad \text{where} \qquad \varphi(r) := \begin{cases} |r| & \text{if } |r| \geqslant 1, \\ \frac{1+r^2}{2} & \text{otherwise.} \end{cases}$$

The function ψ is convex, and there holds $\psi(x,0) = |x|$, while $\psi(0,y) = y^2/2$. One checks that ψ is differentiable away from the origin, but not at $(x,y) = 0$. In fact $\partial\psi(0,0) = \operatorname{conv}\{(-1,0),(1,0)\}$, and elsewhere,

$$\nabla\psi(x,y) = \begin{cases} (\operatorname{sign}(x),0) & \text{if } |x|/y^2 \geqslant 1, \\ \left(\frac{x}{y^2}, y\left(1 - \frac{x^2}{y^4}\right)\right) & \text{otherwise.} \end{cases}$$

Let μ be the normalized Lebesgue measure on $\{\psi \leqslant 1\}$, and $\nu := (\nabla\psi)_\#\mu$. Then ν is supported in the compact convex set $C := \bigcup_{x\in\{\psi\leqslant 1\}} \partial_x\psi$, and $\nabla\psi$ is the unique optimal transport map from μ to ν. On this example, we can compute that ν is the sum of an absolutely continuous measure with density $g(y_1,y_2) = \frac{y_2^2}{(1-y_1^2)^3}$ over C, and of two Dirac masses located as $\pm(1,0)$. We observe that $\operatorname{supp}\nu = C$ and that the density g is not bounded from below.

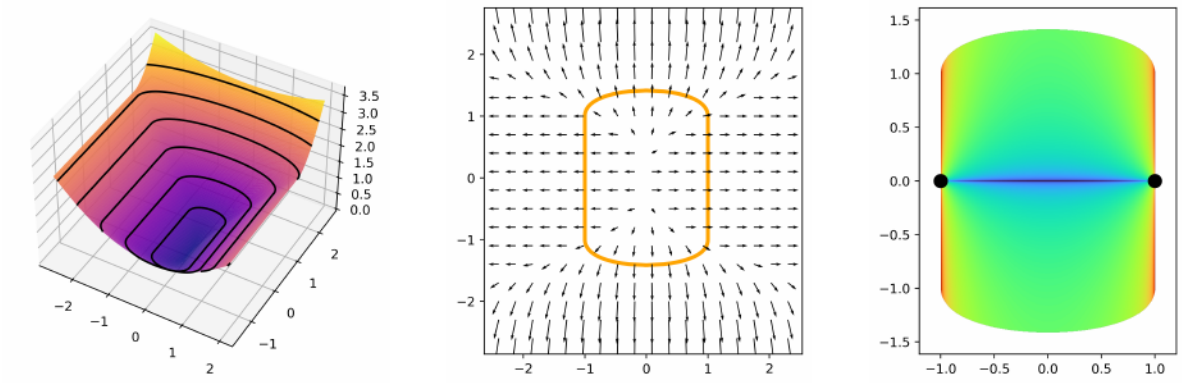


Figure 3: Function ψ, gradient map $\nabla\psi$, and image measure ν.

Level sets of the function ψ are indicated in solid black lines. The gradient field is scaled for visibility, and a level line of the function ψ is plotted in solid line for reference. The image measure has two atoms (black dots) and a density with respect to the Lebesgue measure that explodes as $|y_1| \to 1$. The density degenerates to 0 when $|y_2|$ is close to 0. In all images, the horizontal axis corresponds to abscissa, and the vertical one to ordinate.

Let us point where the reasoning from Lemma 3.3 fails. The only point where the optimal transport map is discontinuous is at $x = (0,0)$, with here $G(x) = [y_0, y_1]$ for $y_0 = (-1,0)$ and $y_1 = (1,0)$. The assumptions of the Lemma are satisfied by the sequence $x_n = (0, t_n)$ for some decreasing sequence $t_n \to 0$, and $y_n = \nabla\psi(x_n) = (0, t_n)$, that converges to $y = (0,0) \in G(x)$. Here the assumption (h3) is trivially satisfied since $\langle x_n - x, y_1 - y_0\rangle = 0$ for any n. To proceed with the reasoning, one would compare

- the measure that ν puts on the triangle $T_n = \mathrm{conv}\{y_{2/3}, y_{3/4}, \overline{y}_n\} = \mathrm{conv}\{(1/3,0),(1/2,0),(1/3,2t_n/3)\}$,
- the measure that μ puts on the region S_n, which we know to be bounded by $c|x_n - x|^2\mathcal{L}(T_n) \simeq t_n^3$.

Here we can explicitly compute that

$$\nu(T_n) = \int_{y_1=1/3}^{1/2} \int_{y_2=0}^{\frac{2t_n}{3}\left(1-\frac{y_1-1/3}{1/2-1/3}\right)} \frac{y_2^2}{(1-y_1^2)^3} dy_2 dy_1 = \int_{y_1=1/3}^{1/2} \frac{\left(\frac{2t_n}{3}\left(1-\frac{y_1-1/3}{1/2-1/3}\right)\right)^3}{3(1-y_1^2)^3} dy_1 = \tilde{c}t_n^3.$$

Therefore, in the inequality (3), the two terms $\nu(T_n)$ and (the bound on) $\mu(S_n)$ are of the same order t_n^3, and the contradiction cannot be carried.

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